A Hahn-Series Coefficientwise Proof of the Hickerson–Mortenson Hecke–Appell Identities

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Abstract

We give a coefficientwise proof guided by an ordered Hahn-series expansion of the Hickerson–Mortenson identity expressing the Hecke-type double sum $f_{n,n+p,n}$ as an Appell–Lerch sum plus a theta correction, for all $n, p \geq 1$. The decomposition itself is derived without convergence estimates or Ramanujan's ${}_1\psi_1$ summation, using a single algebraic mechanism: the Hahn order assigns each Appell–Lerch denominator $1/(1-M)$ a branch sign $\mathrm{sg}(\gamma)$ determined by the monomial order, and the resulting coefficient-level sign-wall kernel sg_2 is the theta correction—not an ad hoc residue, but the forced cost of the chosen expansion chamber. A row-exponent ring identity collapses the multi-index Appell exponent to the Hecke quadratic $H(r,s)$ plus a pure theta shell $N\binom{w+1}{2}$. The remaining wall term is identified with the Hickerson–Mortenson theta correction $\theta_{n,p}$ via a coefficientwise Hahn proof of the Kronecker partial-fraction identity. At the paper level, the identity is a family of $\mathbb{Z}$-coefficient identities, one per monomial. The Lean statements realize these coefficient kernels over $\mathbb{Q}$; integrality and general-ring base change are not separate formalized theorems. The general coefficient theorem, the source-faithful comparison with HM's printed correction, and the coefficientwise Kronecker argument have been verified in Lean 4 with zero `sorry` and no axioms beyond `propext`, `Classical.choice`, and `Quot.sound`.

1 Introduction

Motivation: bilateral algebra needs an order

Naive bilateral formal manipulation is inconsistent. In a Laurent ring, the geometric identity $(1-z)\sum_{k\geq 0} z^k = 1$ is unambiguous, but a bilateral "identity" $(1-z)\sum_{k\in\mathbb{Z}} z^k = 0$ would yield $1 = 0$ by dividing. The root cause is that $\sum_{k\in\mathbb{Z}} z^k$ has no well-ordered support. In a Hahn-series ring with a lexicographic monomial order, the inverse $(1-M)^{-1}$ is uniquely defined: it expands as $\sum_{\ell\geq 0} M^\ell$ or $-\sum_{\ell\geq 1} M^{-\ell}$, depending on the sign of the leading exponent of M. The branch choice is forced, and the branch sign is the price the algebra pays for well-definedness.

This paper shows that the Hickerson–Mortenson theta correction in a Hecke-type identity is exactly that price: the sign-wall kernel sg_2 produced by expanding Appell–Lerch denominators in the Hahn ring.

Background

Hickerson and Mortenson [9] proved that every Hecke-type double sum $f_{n,n+p,n}$ (stated under $\gcd(n,p)=1$) decomposes as an Appell–Lerch sum $g_{n,n+p,n}$ plus a theta correction $\theta_{n,p}/\bar{J}_{0,N}$, where $N = np(2n+p)$. Their proof uses analytic functional equations and a growth-based uniqueness argument. Mortenson [11] gave an alternative proof using Ramanujan's ${}_1\psi_1$ summation. Both proofs rely on convergence in the region $|q| < 1$ and produce the theta correction as an analytic residue.

For background on mock theta functions and Appell–Lerch sums, see Zwegers [17] and Chan [4]; for Bailey-pair approaches to Hecke-type double sums producing one-variable specializations, see Andrews [1] and Lovejoy–Osburn [10].

This paper

We give a Hahn-series coefficientwise proof of the decomposition that replaces the analytic ingredients by formal algebra. The key observation is that the Hahn order assigns each Appell–Lerch denominator $1/(1-M)$ a branch sign $\mathrm{sg}(\gamma)$ determined by the monomial order. After clearing the $\bar{J}$-factor and expanding all Appell rows, the resulting coefficient is a sign-wall function sg_2 that differs from the Hecke cone indicator by exactly the theta correction $\bar{\Theta}_{n,p}$.

The proof reduces to two ingredients:

1. A *row-exponent ring identity* (Proposition 4.3) showing that the Appell row exponent simplifies to $H(r,s) + N\binom{w+1}{2}$.
2. The *signed Hahn geometric inverse* (Lemma 3.3) producing the branch coefficient $(\mathrm{sg}(s) - \mathrm{sg}(V_w))/2$ per row.

The displayed kernels are integer-valued, so the paper argument gives $\mathbb{Z}$-identities one coefficient at a time. The Lean theorem is stated over $\mathbb{Q}$ and does not separately package integrality or base change to an arbitrary commutative ring. The coefficient data is nevertheless available as input to radial-limit and quantum-modular analysis [16, 6], where $|q| < 1$ proofs do not apply directly.

Table 1 summarizes how each analytic ingredient is replaced.

Analytic ingredient	**Hahn-series replacement**
Convergence ($\lvert q\rvert < \lvert x\rvert < 1$)	Lexicographic monomial order
Geometric expansion branch	Signed Hahn inverse (Lemma 3.3)
Analytic continuation	Coefficientwise identity
Poles, residues, uniqueness	Coefficientwise kernel identity
Theta correction formula	Sign-wall boundary term (sg_2)
${}_1\psi_1$ summation	Not used (Appendix uses Kronecker PF)

Table 1: Analytic ingredients and their formal replacements.

Scope. The coefficientwise proof of the sign-wall identity is contained in Sections 2–5. The analytic specialization (recovering HM's analytic identity) requires matching the Hahn branch with the convergent branch on the unit torus (Remark 3.2), plus analytic continuation in x, y. The identification with HM's printed theta correction is proved separately in Appendix A. Its Kronecker step is proved coefficientwise from the signed Hahn inverse and a reflected boundary cancellation. The Lean 4 development verifies this appendix and the full general-(n,p) coefficient assembly; the earlier $f_{2,3,2}$ proof remains as an independent regression instance in HM's spiral parametrization.

Outline. Section 2 gives the definitions and sign conventions. Section 3 proves the Hahn inverse lemma. Section 4 proves the row-coefficient propositions. Section 5 proves the main theorem. Section 6 gives three examples. Section 7 describes the Lean 4 formalization.

2 Definitions

We use the Jacobi theta function $j(z;q) = \sum_{k\in\mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k$, the Appell–Lerch sum $m(x,q,z) = \frac{1}{j(z;q)} \sum_{k\in\mathbb{Z}} \frac{(-1)^k q^{\binom{k}{2}} z^k}{1-q^{k-1}xz}$, the notation $\bar{J}_{0,N} = j(-1;\, q^N)$, and the Euler product $J_m = \prod_{i\geq 1}(1-q^{mi})$.

Definition 2.1 (Hecke-type double sum)**.**

$$f_{A,B,C}(x,y;q) = \Big(\sum_{r,s\geq 0} - \sum_{r,s<0}\Big)(-1)^{r+s} x^r y^s\, q^{A\binom{r}{2}+Brs+C\binom{s}{2}}.$$

For $f_{n,n+p,n}$: set $B = n+p$, $\Delta = p(2n+p)$, $N = n\Delta$, and

$$H(r,s) = n\tbinom{r}{2} + (n+p)rs + n\tbinom{s}{2}.$$

Define the sign-wall functions $\mathrm{sg}(u) = 1$ *if* $u \geq 0$, $\mathrm{sg}(u) = -1$ *if* $u < 0$, *and* $\mathrm{sg}_2(u,v) = (\mathrm{sg}(u) + \mathrm{sg}(v))/2$. *Thus* sg_2 *takes values in* $\{-1,0,1\}$*: it equals* 1 *when both arguments are nonnegative,* -1 *when both are negative, and* 0 *when they disagree in sign.*

$u\backslash v$	$v \geq 0$	$v < 0$
$u \geq 0$	$\mathrm{sg}_2 = 1$	$\mathrm{sg}_2 = 0$
$u < 0$	$\mathrm{sg}_2 = 0$	$\mathrm{sg}_2 = -1$

Definition 2.2 (Appell–Lerch row decomposition)**.** *Set* $A_0 = \frac{np(2n+p+1)}{2}$. *For* $t = 0,\dots,n-1$, *define the s-split rows*

$$G_t^{(s)} = (-y)^t\, q^{n\binom{t}{2}}\, j(q^{Bt}x;\, q^n) \cdot m\Big(-q^{A_0-t\Delta}\frac{(-y)^n}{(-x)^B},\ q^N,\ -1\Big),$$

and the r-split rows

$$G_t^{(r)} = (-x)^t\, q^{n\binom{t}{2}}\, j(q^{Bt}y;\, q^n) \cdot m\Big(-q^{A_0-t\Delta}\frac{(-x)^n}{(-y)^B},\ q^N,\ -1\Big).$$

Define $g_{n,n+p,n} = \sum_{t=0}^{n-1}(G_t^{(s)} + G_t^{(r)})$.

These parameters match [9], Theorem 0.3, specialized to $(z_0,z_1) = (-1,-1)$.

Definition 2.3 (Bar-J-cleared theta correction)**.**

$$\bar{\Theta}_{n,p} = \sum_{r,s,w\in\mathbb{Z}} (-x)^r (-y)^s\ \mathrm{sg}_2\Big(r + nw + \tfrac{n-1}{2},\ s - nw - \tfrac{n+1}{2}\Big)\, q^{H(r,s)+N\binom{w+1}{2}}.$$

Proposition 2.4 (Identification with HM)**.** *With the conventions above,* $\bar{\Theta}_{n,p}$ *and* $\theta_{n,p}^{\mathrm{HM}}$ *have the same coefficient function for all* $n,p \geq 1$. *Here* $\theta_{n,p}^{\mathrm{HM}}$ *is the theta correction in [9], Theorem 0.3 (stated there under* $\gcd(n,p) = 1$*), specialized to* $(z_0,z_1) = (-1,-1)$*, taking HM's printed formula as the definition of* $\theta_{n,p}^{\mathrm{HM}}$ *in the non-coprime case.*

Proof. Appendix A gives the direct coefficient comparison. It uses the coefficientwise Kronecker partial-fraction identity (Lemma A.1) to expand the HM finite theta quotient, then reindexes from the residue pair (r^*,s^*) to the integer triple (R,S,w). The exponent collapses to $H(R,S)+N\binom{w+1}{2}$, and the branch signs produce $\mathrm{sg}_2(U_w,V_w)$.

Note: This identification is logically independent of the main proof (Theorem 2.6). It serves to match $\bar{\Theta}_{n,p}$ against HM's printed formula; the Hecke–Appell decomposition itself (Theorem 2.6) is proved without it. □

Remark 2.5 (Consistency check)**.** *As a consistency check (using HM's theorem, not part of our independent proof), compare coefficients in the cleared HM identity $\bar{J}\cdot f = \bar{J}\cdot g + \theta^{\mathrm{HM}}_{n,p}$ and Theorem 2.6. Subtracting shows that $\bar{\Theta}_{n,p}$ and $\theta^{\mathrm{HM}}_{n,p}$ have the same coefficient function. The self-contained proof of this identification is in Appendix A.*

We record the HM theta correction for completeness. With $r = r^* + \{(n-1)/2\}$, $s = s^* + \{(n-1)/2\}$:

$$\begin{aligned}\theta^{\mathrm{HM}}_{n,p} = \sum_{r^*=0}^{p-1}\sum_{s^*=0}^{p-1} & q^{n\binom{r-(n-1)/2}{2}+B(r-(n-1)/2)(s+(n+1)/2)+n\binom{s+(n+1)/2}{2}} \\ & \times (-x)^{r-(n-1)/2}(-y)^{s+(n+1)/2} \\ & \times \frac{J^3_{p^2(2n+p)}\, j(-q^{np(s-r)}x^n/y^n;\, q^{np^2})\, j(q^{p(2n+p)(r+s)+p(n+p)}x^py^p;\, q^{p^2(2n+p)})}{j(q^{p(2n+p)r+p(n+p)/2}(-y)^B/(-x)^n,\, q^{p(2n+p)s+p(n+p)/2}(-x)^B/(-y)^n;\, q^{p^2(2n+p)})}.\end{aligned}$$

Here $j(a,b;Q) := j(a;Q)\, j(b;Q)$.

Theorem 2.6 (Main coefficient identity)**.** *Let $n, p \geq 1$. For every $(e,r,s) \in \mathbb{Z}^3$,*

$$[q^e x^r y^s]\big(\bar{J}_{0,N}\cdot f_{n,n+p,n}\big) = [q^e x^r y^s]\big(\bar{J}_{0,N}\cdot g_{n,n+p,n} + \bar{\Theta}_{n,p}\big).$$

Remark 2.7. *Neither Theorem 2.6 nor Proposition 2.4 requires* $\gcd(n,p) = 1$*: the proof uses only the division algorithm $s = n\ell + t$ (valid for all $n \geq 1$), polynomial exponent algebra, and the Kronecker partial-fraction identity, none of which involve coprimality. Hickerson–Mortenson [9] state their Theorem 0.3 under* $\gcd(n,p) = 1$*; our results extend the identity to all $n, p \geq 1$ by taking HM's printed formula as the definition of $\theta^{\mathrm{HM}}_{n,p}$ in the non-coprime case.*

3 The Signed Hahn Geometric Inverse

3.1 Coefficient ring

We work in the Hahn-series ring [7] $S = \mathrm{HahnSeries}(\Gamma, \mathbb{Q})$ where $\Gamma = \mathbb{Z}^3$ with the lexicographic order (e,a,b), so that $q = \mathfrak{q}^{(1,0,0)}$, $x = \mathfrak{q}^{(0,1,0)}$, $y = \mathfrak{q}^{(0,0,1)}$. A monomial $q^e x^a y^b$ has leading exponent (e,a,b), and the sign $\mathrm{sg}(\gamma)$ refers to the lexicographic sign of γ. In particular, for the Appell denominator monomial $M_{k,t} = q^D \cdot (-y)^n/(-x)^B$, the leading exponent is $(D,-B,n)$, which is positive iff $D > 0$ (since D is the first component).

All identities below are interpreted coefficientwise: for each fixed (e,r,s), the coefficient $[q^e x^r y^s]$ of every displayed series is a finite sum. The Hahn order is used only to define the branch of each geometric inverse (Lemma 3.3). Lemma 4.1 below ensures that the theta-shell sums are locally finite. More precisely, for each displayed bilateral sum $\sum_k q^{N\binom{k}{2}}/(1-M_{k,t})$, the monomial forcing (Step 1 in Proposition 4.4) pins (t,ℓ,u) from (r,s), and e determines k to at most two values, so only finitely many terms contribute to each coefficient.

Remark 3.1 (Coefficientwise reading)**.** *Theorem 2.6 is an identity of coefficient functions: for each fixed $(e,r,s) \in \mathbb{Z}^3$, both displayed coefficients are finite sums and agree. The Hahn order determines the branch of each individual geometric inverse (Lemma 3.3); Lemma 4.1 and Proposition 4.4 establish the local finiteness needed for the coefficient calculation. We do not infer global well-ordered support from this local statement, and this paper does not supply a Hahn-series wrapper for the assembled three-variable families. In particular, the occurrence of negative q-powers by itself would not decide Hahn-series membership.*

Remark 3.2 (Analytic specialization)**.** *The formal identity specializes to HM's analytic identity in any chamber where $|M_{k,t}| < 1$ iff $\gamma_{k,t} > 0$. Since the denominator monomial $M_{k,t}$ has leading q-exponent $\gamma_{k,t}$ and auxiliary-variable factor $(-y)^n/(-x)^B$, this condition holds on the unit torus $|x| = |y| = 1$ (avoiding Appell poles): there $|M_{k,t}| = |q|^{\gamma_{k,t}}$, so the Hahn branch coincides with the convergent branch and the identity specializes by absolute convergence. For generic $|x|, |y| \neq 1$, the leading q-exponent may not dominate $|M_{k,t}|$, and analytic continuation in x, y is required to recover HM's identity over all of $\{|q| < 1\} \times (\mathbb{C}^*)^2$.*

Lemma 3.3 (Hahn geometric inverse)**.** *Let Γ be a linearly ordered abelian group and let $S = \mathrm{HahnSeries}(\Gamma, R)$ be the ring of Hahn series with well-ordered support. For a unit monomial $M = c\,\mathfrak{q}^\gamma \in S$ with $c \in R^\times$ and $\gamma \neq 0$:*

$$(1-M)^{-1} = \begin{cases} \displaystyle\sum_{\ell \geq 0} M^\ell, & \gamma > 0, \\ \displaystyle -\sum_{\ell \geq 1} M^{-\ell}, & \gamma < 0. \end{cases}$$

The coefficient of M^ℓ is therefore 1 for $\ell \geq 0$ when $\gamma > 0$, and -1 for $\ell \leq -1$ when $\gamma < 0$; zero otherwise.

Proof. For $\gamma > 0$, the support $\{0, \gamma, 2\gamma, \ldots\}$ is well-ordered in Γ, so the series lies in S. For $\gamma < 0$, factor $(1-M)^{-1} = -M^{-1}(1-M^{-1})^{-1}$; since $-\gamma > 0$, the factor $(1-M^{-1})^{-1}$ has well-ordered support, and the product lies in S. □

Lemma 3.4 (Branch coefficient)**.** *For a unit monomial M with leading exponent $\gamma \neq 0$ in the linearly ordered group, the coefficient of M^ℓ in $(1-M)^{-1}$ is*

$$[M^\ell](1-M)^{-1} = \frac{\mathrm{sg}(\ell) + \mathrm{sg}(\gamma)}{2},$$

where we use the convention $\mathrm{sg}(0) = 1$.

Proof. By Lemma 3.3: if $\gamma > 0$, the coefficient is 1 for $\ell \geq 0$ and 0 for $\ell < 0$, which equals $(1 + \mathrm{sg}(\ell))/2 = (\mathrm{sg}(\ell) + \mathrm{sg}(\gamma))/2$. If $\gamma < 0$, the coefficient is -1 for $\ell \leq -1$ and 0 for $\ell \geq 0$, which equals $(-1 + \mathrm{sg}(\ell))/2 = (\mathrm{sg}(\ell) + \mathrm{sg}(\gamma))/2$. □

Lemma 3.5 (Cleared Appell numerator)**.** $\bar{J}_{0,N}\, m(a, q^N, -1) = \sum_{k \in \mathbb{Z}} q^{N\binom{k}{2}}/(1 + q^{N(k-1)}a)$.

Proof. With $z = -1$, the denominator of $m(a, q^N, -1)$ is $j(-1; q^N) = \bar{J}_{0,N}$, so the factor cancels. In the numerator, $(-1)^k \cdot (-1)^k = 1$, and $1 - q^{N(k-1)}a \cdot (-1) = 1 + q^{N(k-1)}a$, giving the stated bilateral sum. □

4 Row Coefficients

Lemma 4.1 (Local finiteness)**.** *For fixed (e, r, s), the equation $e = H(r,s) + N\binom{w+1}{2}$ has at most two solutions $w \in \mathbb{Z}$ (namely w and $-w-1$ when $e - H(r,s) = N\binom{w+1}{2}$). Hence every coefficient extraction in the proof is a finite sum.*

Proof. $\binom{w+1}{2} = (e - H(r,s))/N$ determines $w(w+1)/2$, which has at most two integer solutions. □

Lemma 4.2 (Branch sign)**.** *Let $0 \le t < n$, $k \in \mathbb{Z}$, and set*

$$W_{k,t} = t + \tfrac{n-1}{2} - nk, \qquad \gamma_{k,t} = N(k-1) + A_0 - t\Delta.$$

Then $\gamma_{k,t} \neq 0$ and $\operatorname{sg}(\gamma_{k,t}) = -\operatorname{sg}(W_{k,t})$.

Proof. Write $A = nk - t \in \mathbb{Z}$, so $W_{k,t} = (n-1)/2 - A$ and $\gamma_{k,t} = \Delta(A - n/2) + np/2$.

If $W_{k,t} \ge 0$: then $A \le (n-1)/2$, so $\gamma_{k,t} \le -p(n+p)/2 < 0$, so $\operatorname{sg}(\gamma_{k,t}) = -1 = -\operatorname{sg}(W_{k,t})$.

If $W_{k,t} < 0$: then $A > (n-1)/2$. Since $A \in \mathbb{Z}$: for $n = 2m$, $A \ge m$, so $\gamma_{k,t} \ge np/2 > 0$; for $n = 2m+1$, $A \ge m+1$, so $\gamma_{k,t} \ge \Delta/2 + np/2 > 0$. In both cases $\operatorname{sg}(\gamma_{k,t}) = 1 = -\operatorname{sg}(W_{k,t})$. □

Proposition 4.3 (Row exponent identity)**.** *For $s = n\ell + t$ with $0 \le t < n$:*

$$n\tbinom{t}{2} + n\tbinom{r+B\ell}{2} + Bt(r + B\ell) + N\tbinom{k}{2} + \ell\big(N(k-1) + A_0 - t\Delta\big) = H(r,s) + N\tbinom{k+\ell}{2}.$$

Proof. Both sides are polynomials in (r, t, ℓ, k, n, p) of degree 3. Substituting $s = n\ell + t$, $B = n + p$, $\Delta = B^2 - n^2 = p(2n+p)$, $N = n\Delta$, $A_0 = np(2n+p+1)/2$, and expanding both sides, the difference vanishes identically.[1] The same identity holds with r and s interchanged. □

Proposition 4.4 (s-row coefficient)**.** *Let $V_w = s - nw - (n+1)/2$. Then*

$$[q^e\, x^r\, y^s]\ \bar{J}_{0,N} \sum_{t=0}^{n-1} G_t^{(s)} = (-1)^{r+s} \sum_{w \in \mathbb{Z}} \frac{\operatorname{sg}(s) - \operatorname{sg}(V_w)}{2}\, \mathbf{1}_{e = H(r,s) + N\binom{w+1}{2}}.$$

Proof. Write $s = n\ell + t$ with $0 \le t < n$. The Appell parameter is $a_t = -q^{A_0 - t\Delta}(-y)^n/(-x)^B$. By Lemma 3.5,

$$\bar{J}_{0,N} G_t^{(s)} = (-y)^t q^{n\binom{t}{2}} \sum_{u \in \mathbb{Z}} (-1)^u q^{n\binom{u}{2} + Btu}\, x^u \sum_{k \in \mathbb{Z}} \frac{q^{N\binom{k}{2}}}{1 - M_{k,t}}, \tag{1}$$

where

$$M_{k,t} = q^{N(k-1) + A_0 - t\Delta}\, \frac{(-y)^n}{(-x)^B}$$

is the denominator monomial (note: $1 + q^{N(k-1)} a_t = 1 - M_{k,t}$).

Step 1: monomial forcing. The monomial $x^r y^s$ from $x^u \cdot M_{k,t}^\ell$ forces $u - B\ell = r$ (so $u = r + B\ell$) and $n\ell + t = s$ (so $\ell = (s-t)/n$). Since $M_{k,t}^\ell = q^{\ell\gamma_{k,t}}(-y)^{n\ell}(-x)^{-B\ell}$, the total sign is $(-1)^t \cdot (-1)^u \cdot (-1)^{n\ell} \cdot (-1)^{-B\ell} = (-1)^{t+u+n\ell-B\ell} = (-1)^{t+(r+B\ell)+n\ell-B\ell} = (-1)^{r+t+n\ell} = (-1)^{r+s}$ (using $t + n\ell = s$).

Step 2: exponent. By Proposition 4.3, the total q-exponent is $H(r,s) + N\binom{k+\ell}{2}$. Set $w = k + \ell - 1$; then $\binom{k+\ell}{2} = \binom{w+1}{2}$.

Step 3: branch coefficient. The leading exponent of $M_{k,t}$ in the lexicographic Hahn order (with q first) is $\gamma_{k,t} = N(k-1) + A_0 - t\Delta$. By Lemma 3.4, the coefficient of $M_{k,t}^\ell$ in $(1 - M_{k,t})^{-1}$ is $(\operatorname{sg}(\ell) + \operatorname{sg}(\gamma_{k,t}))/2$.

After substitution $k = w - \ell + 1$:

$$\gamma_{k,t} = N(w - \ell) + A_0 - t\Delta.$$

By Lemma 4.2 (with $W_{k,t} = V_w$), $\operatorname{sg}(\gamma_{k,t}) = -\operatorname{sg}(V_w)$. Therefore the branch coefficient is $(\operatorname{sg}(\ell) - \operatorname{sg}(V_w))/2$. Since $s = n\ell + t$ with $0 \le t < n$ and $\operatorname{sg}(0) = 1$, one has $\operatorname{sg}(\ell) = \operatorname{sg}(s)$, giving the stated formula. □

[1] A Sage script verifying this polynomial identity and the appendix exponent identity (A.15) is included as `check_exponents.sage`.

Proposition 4.5 (r-row coefficient)**.** *Let $U_w = r + nw + (n-1)/2$. Then*

$$[q^e\, x^r\, y^s]\; \bar{J}_{0,N} \sum_{t=0}^{n-1} G_t^{(r)} = (-1)^{r+s} \sum_{w\in\mathbb{Z}} \frac{\operatorname{sg}(r) - \operatorname{sg}(U_w)}{2}\, \mathbf{1}_{e=H(r,s)+N\binom{w+1}{2}}.$$

Proof. Write $r = n\ell + t$ with $0 \le t < n$. The Appell parameter is $a_t = -q^{A_0 - t\Delta}(-x)^n/(-y)^B$. By Lemma 3.5,

$$\bar{J}_{0,N} G_t^{(r)} = (-x)^t q^{n\binom{t}{2}} \sum_{v\in\mathbb{Z}} (-1)^v q^{n\binom{v}{2}+Btv}\, y^v \sum_{k\in\mathbb{Z}} \frac{q^{N\binom{k}{2}}}{1 - M'_{k,t}},$$

where $M'_{k,t} = q^{N(k-1)+A_0-t\Delta}(-x)^n/(-y)^B$.

Monomial forcing: $y^v \cdot (M'_{k,t})^\ell$ gives $v - B\ell = s$ and $n\ell + t = r$. Total sign: $(-1)^{t+v+n\ell-B\ell} = (-1)^{s+t+n\ell} = (-1)^{r+s}$.

Exponent: by the symmetric row-exponent identity (Proposition 4.3 with $r \leftrightarrow s$), the q-exponent is $H(r,s) + N\binom{k+\ell}{2}$. Set $w = -k - \ell$ (the opposite sign convention from the s-row, chosen so that U_w matches $W_{k,t}$); then $\binom{k+\ell}{2} = \binom{-w}{2} = \binom{w+1}{2}$ and $U_w = r + nw + (n-1)/2 = t + (n-1)/2 - nk$.

Branch sign: by Lemma 4.2 with $W_{k,t} = U_w$, $\operatorname{sg}(\gamma'_{k,t}) = -\operatorname{sg}(U_w)$. The branch coefficient is $(\operatorname{sg}(\ell) - \operatorname{sg}(U_w))/2 = (\operatorname{sg}(r) - \operatorname{sg}(U_w))/2$. □

Corollary 4.6 (Cleared Appell coefficient)**.**

$$[q^e\, x^r\, y^s]\; \bar{J}_{0,N}\, g_{n,n+p,n} = (-1)^{r+s} \sum_{w\in\mathbb{Z}} \big(\operatorname{sg}_2(r,s) - \operatorname{sg}_2(U_w, V_w)\big)\, \mathbf{1}_{e=H(r,s)+N\binom{w+1}{2}}.$$

Proof. Add Propositions 4.4 and 4.5 and use $\frac{\operatorname{sg}(r)-\operatorname{sg}(U_w)}{2} + \frac{\operatorname{sg}(s)-\operatorname{sg}(V_w)}{2} = \operatorname{sg}_2(r,s) - \operatorname{sg}_2(U_w, V_w)$. □

5 Proof of the Main Identity

Proof of Theorem 2.6. Fix $e, r, s \in \mathbb{Z}$. Since $\bar{J}_{0,N} = \sum_{w\in\mathbb{Z}} q^{N\binom{w+1}{2}}$, the Hecke coefficient is

$$[q^e\, x^r\, y^s]\; \bar{J}_{0,N}\, f_{n,n+p,n} = (-1)^{r+s} \sum_{w\in\mathbb{Z}} \operatorname{sg}_2(r,s)\, \mathbf{1}_{e=H(r,s)+N\binom{w+1}{2}}.$$

By Corollary 4.6:

$$[q^e\, x^r\, y^s]\; \bar{J}_{0,N}\, g_{n,n+p,n} = (-1)^{r+s} \sum_{w\in\mathbb{Z}} \big(\operatorname{sg}_2(r,s) - \operatorname{sg}_2(U_w, V_w)\big)\, \mathbf{1}_{e=H(r,s)+N\binom{w+1}{2}}.$$

Subtracting:

$$[q^e\, x^r\, y^s]\big(\bar{J}_{0,N}\, f - \bar{J}_{0,N}\, g\big) = (-1)^{r+s} \sum_{w\in\mathbb{Z}} \operatorname{sg}_2(U_w, V_w)\, \mathbf{1}_{e=H(r,s)+N\binom{w+1}{2}},$$

which equals $[q^e\, x^r\, y^s]\, \bar{\Theta}_{n,p}$ by definition. □

6 Examples

6.1 $f_{2,3,2}$ ($n = 2$, $p = 1$, $\Delta = 5$)

Here $N = 10$, $B = 3$, and the Appell sum has $2n = 4$ rows. The main identity gives $\bar{J}_{0,10}\, f_{2,3,2} = \bar{J}_{0,10}\, g_{2,3,2} + \bar{\Theta}_{2,1}$. In the Lean 4 formalization, the right-hand side is packaged as $\mathrm{Ghat}(1) = g + \theta/\bar{J}$, and the identity $f_{2,3,2} = \mathrm{Ghat}(1)$ is proved with zero `sorry`.

6.2 $f_{1,3,1}$ ($n = 1$, $p = 2$, $\Delta = 8$)

Here $N = 8$ and the Appell sum has 2 rows. The cleared theta correction $\bar{\Theta}_{1,2}$ is nonzero. For instance, at $(r, s) = (1, 1)$: $U_0 = 1$, $V_0 = 0$, $U_{-1} = 0$, $V_{-1} = 1$, giving $[q^3\, xy]\, \bar{J}_{0,8}\, f_{1,3,1} = 2$, $[q^3\, xy]\, \bar{J}_{0,8}\, g_{1,3,1} = 0$, and $[q^3\, xy]\, \bar{\Theta}_{1,2} = \mathrm{sg}_2(1, 0) + \mathrm{sg}_2(0, 1) = 2$.

6.3 $f_{1,4,1}$ ($n = 1$, $p = 3$, $\Delta = 15$)

Here $N = 15$ and the Appell sum has 2 rows. The direct coefficient proof applies without modification. The correction $\bar{\Theta}_{1,3}$ is nonzero: at $(r, s) = (1, 1)$, $[q^4\, xy]\, \bar{\Theta}_{1,3} = 2$, matching the defect $[q^4\, xy](\bar{J} \cdot f - \bar{J} \cdot g) = 2 - 0 = 2$.

6.4 $f_{2,4,2}$ ($n = 2$, $p = 2$, $\gcd = 2$)

Here $\gcd(n, p) = 2$, $N = 24$, $B = 4$, $\Delta = 12$, and the Appell sum has $2n = 4$ rows. By the q-scaling identity (Section 8), $f_{2,4,2}(x, y, q) = f_{1,2,1}(x, y, q^2)$, so the Hecke sum admits both the $(2, 2)$-decomposition of Theorem 2.6 and the scaled primitive $(1, 1)$-decomposition. The coefficient proof applies to the $(2, 2)$-decomposition directly: no step uses $\gcd(n, p) = 1$.

7 Formal Verification

Table 2 summarizes the Lean 4 formalization status.

Claim	**Status**	**File / theorem**
$f_{2,3,2}$ (spiral param.)	fully verified	`Chapter10_M3_AppellCoeffNF`
Row-exponent identity	verified $\forall n, p$	`Chapter10_RootPairGeneral`
Branch-sign lemma	verified $\forall n, p$	`Chapter10_RootPairGeneral`
Parity criterion	verified $\forall n, p$	`Chapter10_RootPairGeneral`
Props. 4.4,4.5 (general)	fully verified	`Chapter10_GeneralHahnCoeff`
Main identity, all $n, p \geq 1$	fully verified	`general_boundary_cleared_hm_coefficient`
Kronecker coefficient proof	fully verified	`Chapter10_KroneckerCoeff`
HM residue comparison	fully verified	`Chapter10_HMComparison`

Table 2: Lean 4 formalization coverage. All verified items have zero `sorry` and no axioms beyond `propext`, `Classical.choice`, `Quot.sound`.

The $f_{2,3,2}$ verification (commit `5ead660`) proves the identity `Fhm_eq_Ghat_via_uvOmega` via a wall-coordinate specialization: parity split of the 4 Appell rows, finsum reindex via coordinate equivalences, wall-defect decomposition, and root-pair cancellation. The Lean object `Ghat` is defined as a four-row Appell sum specialized to HM's monomial-spiral parameter (cf. [9], Theorem 0.6). This older regression uses a different specialization interface and is logically independent of the general coefficient theorem formalized here.

The general algebraic core is formalized in `Chapter10_RootPairGeneral.lean`; `Chapter10_GeneralHahnCoeff.lean` adds the explicit finite residue rows, their monomial-forcing collapse, local finiteness, both row formulas, and the main coefficient assembly. The appendix is formalized independently: `Chapter10_KroneckerCoeff.lean` proves the PF quasi-period by an inverse-coefficient telescope, and `Chapter10_HMComparison.lean` retains both residue variables and all three bilateral indices in a constrained source type, proves its equivalence with the single shell

index w, and checks the branch-sign and exponent formulas (A.10)–(A.17). The final module is `Chapter10_HahnHeckeAppell`; its main theorem is `general_boundary_cleared_hm_coefficient`.

The self-contained fixed registry source, its exact proof closure, and the paper snapshot are archived together at `https://github.com/xiangyazi24/palomar-hahn-hecke-appell`. The relevant Lean files are in the `Pending/` directory and the final audit target is `QseriesFormalization.Pending.Chapter10_HahnHeckeAppellAudit`. A Sage script verifying both the row-exponent identity (Proposition 4.3) and the appendix exponent identity (A.15) is included as `check_exponents.sage`.

8 Discussion

Comparison with existing proofs

The Hickerson–Mortenson proof [9] uses analytic uniqueness; Mortenson [11] uses ${}_1\psi_1$. Our proof replaces both with Hahn-series coefficient algebra. The sg_2 structure arises from the Hahn geometric inverse expansion direction, providing a formal-algebraic interpretation of the Appell–Lerch sign-wall coefficients.

The row-exponent identity (Proposition 4.3) is the algebraic heart: it collapses the multi-index Appell row exponent into the Hecke exponent $H(r,s)$ plus a theta shell $N\binom{w+1}{2}$. This identity is verifiable by `ring` and is the key step that makes the coefficient comparison self-contained.

We note the relationship with Bailey-pair approaches to Hecke-type double sums [1, 10]. Those methods are also formal-series arguments and produce identities for one-variable specializations; however, they do not establish the two-variable (x,y)-generic Appell–Lerch *decomposition*, where the bilateral Appell parameter is precisely where analytic or Hahn-algebraic machinery enters.

Non-coprime cases

For $\gcd(n,p) = d > 1$, one has $f_{dn_0,d(n_0+p_0),dn_0}(x,y;q) = f_{n_0,n_0+p_0,n_0}(x,y;q^d)$, so the Hecke sum admits both the direct (n,p)-decomposition (Theorem 2.6) and the scaled primitive (n_0,p_0)-decomposition. The two decompositions have different numbers of Appell rows and different base parameters; their coexistence is a consistency check, not a derivation of one from the other.

Connection to indefinite theta functions and wall-crossing (formal analogy, not used in the proof)

The sg_2 kernel appearing in $\bar{\Theta}_{n,p}$ is of the same type as the sign-wall functions arising in Zwegers' indefinite theta series [17] and in the wall-crossing analysis of Dabholkar–Murthy–Zagier [5]. The connection is structural: the Hecke exponent $H(r,s)$ is an indefinite quadratic form of type $(1,1)$—the same structure that Hecke [8] studied in his original construction of modular forms from indefinite quadratic forms—and the chosen Hahn lexicographic order is the formal expansion chamber analogous to the analytic chamber in which the corresponding monomials satisfy $|q| < 1$.

From a polyhedral viewpoint, the signed Hahn geometric inverse (Lemma 3.3) is a one-dimensional instance of the polarized-cone expansion of Varchenko [12] and Brion [3]: expanding $1/(1-M)$ in one Hahn chamber selects one ray of a one-dimensional cone, and the branch sign records the orientation. The theta correction $\bar{\Theta}_{n,p}$ is then the boundary term arising from the polarization. The use of ordered formal series rings to select expansion branches is also developed by Xin [15] (Malcev–Neumann series and formal residues) and Aparicio Monforte–Kauers [2] (formal Laurent series in several variables).

What distinguishes the formal setting is that the sg_2 kernel remains exact—no Vignéras error function [13] or modular completion is needed, because the Hahn ring has no $\bar{\tau}$ variable to smooth. This connection is included as context; the paper does not prove a modular completion theorem.

Coefficient data for quantum-modular settings (possible application, not proved here)

The paper formulas are integer-valued coefficient identities. The current Lean wrapper checks them over $\mathbb{Q}$ and does not separately formalize integrality or arbitrary-ring base change. The resulting coefficient data is available as input to quantum-modular and radial-limit analysis [16, 6], where the analytic and ${}_1\psi_1$ proofs do not directly supply such data.

Extensions

The approach may extend to other families of indefinite theta identities. For $f_{a,b,c}$ with $a \neq c$ (HM Theorem 0.4), the row-exponent identity generalizes, but the branch-sign lemma requires a case analysis whose boundary cases encode the additional correction structure. The key requirement is a row-exponent collapse analogous to Proposition 4.3; when such a collapse exists, the branch-sign mechanism produces the correction term automatically.

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A Coefficient comparison with the Hickerson–Mortenson theta term

We keep the conventions of the paper:

$$C_2(z) = \binom{z}{2} = \frac{z(z-1)}{2}, \qquad B = n+p, \qquad \Delta = p(2n+p), \qquad N = n\Delta,$$

$$A_0 = \frac{np(2n+p+1)}{2}, \qquad H(R,S) = nC_2(R) + BRS + nC_2(S), \qquad \mathrm{sg}_2(u,v) = \frac{\mathrm{sg}(u)+\mathrm{sg}(v)}{2},$$

and

$$j(z;q) = \sum_{m\in\mathbb{Z}} (-1)^m q^{C_2(m)} z^m, \qquad \mathrm{sg}(0) = 1.$$

Let

$$\alpha = \frac{n-1}{2}, \qquad \beta = \frac{n+1}{2}, \qquad \varepsilon = \{\alpha\} \in \{0, 1/2\}.$$

For $r^*, s^* \in \{0, \dots, p-1\}$, put

$$\widetilde{r} = r^* + \varepsilon, \qquad \widetilde{s} = s^* + \varepsilon, \qquad \rho = \widetilde{r} - \alpha, \qquad \sigma = \widetilde{s} + \beta.$$

Then $\rho, \sigma \in \mathbb{Z}$, and $\widetilde{r}, \widetilde{s} \in \varepsilon + \{0, \dots, p-1\} \subset [0, p)$. Also

$$\widetilde{s} - \widetilde{r} = \sigma - \rho - n.$$

The Hickerson–Mortenson theta correction, specialized to $(z_0, z_1) = (-1, -1)$, can be rewritten as

$$\theta_{n,p}^{\mathrm{HM}} = \sum_{r^*=0}^{p-1} \sum_{s^*=0}^{p-1} q^{H(\rho,\sigma)} (-x)^{\rho} (-y)^{\sigma} j\left(-q^{np(\sigma-\rho-n)} \frac{x^n}{y^n}; q^{np^2}\right) \times \frac{J_Q^3\, j(d_1 d_2; Q)}{j(d_1; Q) j(d_2; Q)}, \tag{A.1}$$

where

$$Q = q^{p\Delta} = q^{p^2(2n+p)},$$

$$d_1 = q^{\Delta\rho + A_0 - np} \frac{(-y)^B}{(-x)^n}, \qquad d_2 = q^{\Delta\sigma - A_0} \frac{(-x)^B}{(-y)^n}.$$

Indeed,

$$d_1 d_2 = q^{\Delta(\rho+\sigma) - np} (-x)^p (-y)^p,$$

which is the second numerator theta argument in HM's formula, since

$$\Delta(\widetilde{r} + \widetilde{s}) + pB = \Delta(\rho + \sigma - 1) + pB = \Delta(\rho + \sigma) - np.$$

Lemma A.1 (Kronecker partial fraction)**.** *With* $j(z; Q) = \sum_m (-1)^m Q^{\binom{m}{2}} z^m$,

$$\frac{J_Q^3\, j(ab; Q)}{j(a; Q)\, j(b; Q)} = \sum_{\eta \in \mathbb{Z}} \frac{b^\eta}{1 - aQ^\eta}. \tag{A.2}$$

Proof. We give the coefficientwise proof used in the Lean development. Put

$$\mathrm{PF}(a) = \sum_{k \in \mathbb{Z}} \frac{(-1)^k Q^{\binom{k+1}{2}}}{1 - aQ^k},$$

with every denominator expanded in its signed Q-first Hahn branch. The coefficientwise partial-fraction theorem

$$j(a; Q)\, \mathrm{PF}(a) = J_Q^3$$

follows by a geometric telescope: all nonconstant a-coefficients cancel, while the constant coefficient is Jacobi's cube identity.

Let $K(a, b) = \sum_{\eta \in \mathbb{Z}} b^\eta / (1 - aQ^\eta)$. The coefficient of b^B in $j(b; Q) K(a, b)$ is

$$\sum_{s \in \mathbb{Z}} \frac{(-1)^s Q^{\binom{s}{2}}}{1 - aQ^{B-s}} = \mathrm{PF}(aQ^B),$$

where the second equality uses $k = -s$ and $\binom{s}{2} = \binom{k+1}{2}$. The inverse-coefficient telescope also gives

$$\mathrm{PF}(aQ^B) = (-1)^B Q^{\binom{B}{2}} a^B\, \mathrm{PF}(a) \qquad (B \in \mathbb{Z}).$$

In the one-step relation, the only boundary term is a coefficient of $j(1; Q)$, whose two roots k and $-k - 1$ cancel. The last display is exactly the coefficient of b^B in $j(ab; Q)\, \mathrm{PF}(a)$. Hence

$$j(b; Q) K(a, b) = j(ab; Q)\, \mathrm{PF}(a)$$

coefficientwise. Multiplication by $j(a;Q)$ and the partial-fraction theorem above gives

$$j(a;Q)j(b;Q)K(a,b) = J_Q^3 j(ab;Q);$$

cancellation in the corresponding Hahn fraction field proves (A.2). This is the classical Kronecker identity (cf. Weil [14], Chapter IV, and [9], Proposition 5.7 and Lemma 5.10), but no residue or Liouville argument is needed here. □

Applying Lemma A.1 with $a = d_1$, $b = d_2$ gives

$$\frac{J_Q^3 j(d_1 d_2;Q)}{j(d_1;Q)j(d_2;Q)} = \sum_{\eta\in\mathbb{Z}} \frac{d_2^{\eta}}{1-Q^{\eta}d_1}. \tag{A.3}$$

Next expand the first numerator theta factor by the Jacobi triple product:

$$\begin{aligned} j\left(-q^{np(\sigma-\rho-n)}\frac{x^n}{y^n};q^{np^2}\right) &= \sum_{\iota\in\mathbb{Z}} (-1)^{\iota} q^{np^2 C_2(\iota)} \left(-q^{np(\sigma-\rho-n)}\frac{x^n}{y^n}\right)^{\iota} \\ &= \sum_{\iota\in\mathbb{Z}} q^{np^2C_2(\iota)+np(\sigma-\rho-n)\iota}(-x)^{n\iota}(-y)^{-n\iota}. \end{aligned} \tag{A.4}$$

Here the signs cancel because $(-1)^{\iota}(-1)^{\iota} = 1$.

For the remaining denominator, set

$$M_{\rho,\eta} = Q^{\eta}d_1 = q^{p\Delta\eta+\Delta\rho+A_0-np}\frac{(-y)^B}{(-x)^n}.$$

Its leading q-exponent is

$$\Gamma_{\rho,\eta} = p\Delta\eta + \Delta\rho + A_0 - np.$$

Formal–analytic bridge. On the unit torus $|x| = |y| = 1$ with $|q| < 1$, $|M_{\rho,\eta}| = |q|^{\Gamma_{\rho,\eta}}$, so $|M_{\rho,\eta}| < 1$ iff $\Gamma_{\rho,\eta} > 0$. Hence the Hahn branch of $(1-M_{\rho,\eta})^{-1}$ coincides with the convergent geometric series. Thus the coefficientwise proof of Lemma A.1 specializes to the analytic quotient on this torus, with the same Hahn branch convention. The signed Hahn geometric expansion gives

$$(1-M_{\rho,\eta})^{-1} = \sum_{\ell\in\mathbb{Z}} c_{\rho,\eta}(\ell) M_{\rho,\eta}^{\ell}, \qquad c_{\rho,\eta}(\ell) = \frac{\mathrm{sg}(\ell)+\mathrm{sg}(\Gamma_{\rho,\eta})}{2}. \tag{A.5}$$

Combining (A.1)–(A.5), a term indexed by $(\rho,\sigma,\iota,\eta,\ell)$ has monomial part

$$(-x)^{\rho+n\iota+B\eta-n\ell}(-y)^{\sigma-n\iota-n\eta+B\ell}.$$

Thus the coefficient of $(-x)^R(-y)^S$ is constrained by

$$R = \rho + n\iota + B\eta - n\ell, \qquad S = \sigma - n\iota - n\eta + B\ell. \tag{A.6}$$

Define

$$w = \ell - \iota - \eta. \tag{A.7}$$

Then (A.6) implies

$$U_w := R + nw + \alpha = \rho + p\eta + \alpha = \widetilde{r} + p\eta, \tag{A.8}$$

$$V_w := S - nw - \beta = \sigma - \beta + p\ell = \widetilde{s} + p\ell. \tag{A.9}$$

Conversely, for fixed R, S, w, equations (A.8)–(A.9) determine the contributing residue pair and summation indices uniquely. Indeed, since $\widetilde{r} \in \varepsilon + \{0, \dots, p-1\}$, there is a unique $\widetilde{r}$ in that residue interval and a unique $\eta \in \mathbb{Z}$ such that

$$R + nw + \alpha = \widetilde{r} + p\eta.$$

Similarly, there is a unique $\widetilde{s} \in \varepsilon + \{0, \dots, p-1\}$ and a unique $\ell \in \mathbb{Z}$ such that

$$S - nw - \beta = \widetilde{s} + p\ell.$$

Then $\rho = \widetilde{r} - \alpha$, $\sigma = \widetilde{s} + \beta$, and (A.7) determines the unique remaining index

$$\iota = \ell - \eta - w.$$

Thus coefficient extraction over the HM residue variables and the bilateral indices is equivalent to summing over $w \in \mathbb{Z}$.

We now identify the branch coefficient. Since $\widetilde{s} \in [0, p)$, equation (A.9) gives

$$\mathrm{sg}(\ell) = \mathrm{sg}(\widetilde{s} + p\ell) = \mathrm{sg}(V_w). \tag{A.10}$$

Indeed this is immediate for $\ell \geq 0$, while for $\ell \leq -1$ one has $\widetilde{s} + p\ell < 0$. Likewise, using (A.8),

$$\begin{aligned} \Gamma_{\rho,\eta} &= p\Delta\eta + \Delta\rho + A_0 - np \\ &= \Delta(\rho + p\eta) + A_0 - np \\ &= \Delta(U_w - \alpha) + A_0 - np \\ &= \Delta U_w + \frac{pB}{2}. \end{aligned} \tag{A.11}$$

If $U_w \geq 0$, then $\Gamma_{\rho,\eta} > 0$. If $U_w < 0$, then $U_w = \widetilde{r} + p\eta$ with $\eta \leq -1$, hence $U_w \leq \widetilde{r} - p \leq -1/2$, and therefore

$$\Gamma_{\rho,\eta} \leq -\frac{\Delta}{2} + \frac{pB}{2} = -\frac{np}{2} < 0.$$

Thus

$$\mathrm{sg}(\Gamma_{\rho,\eta}) = \mathrm{sg}(U_w). \tag{A.12}$$

Combining (A.5), (A.10), and (A.12), the branch coefficient becomes

$$c_{\rho,\eta}(\ell) = \frac{\mathrm{sg}(V_w) + \mathrm{sg}(U_w)}{2} = \mathrm{sg}_2(U_w, V_w). \tag{A.13}$$

It remains to compute the q-exponent. The total exponent of the term indexed by $(\rho, \sigma, \iota, \eta, \ell)$ is

$$\begin{aligned} \mathcal{E} =& H(\rho, \sigma) + np^2 C_2(\iota) + np(\sigma - \rho - n)\iota \\ &+ (\Delta\sigma - A_0)\eta + (p\Delta\eta + \Delta\rho + A_0 - np)\ell. \end{aligned} \tag{A.14}$$

Using (A.6) and (A.7), a direct expansion gives

$$\mathcal{E} = H(R, S) + NC_2(w + 1). \tag{A.15}$$

The following Sage snippet is included as an audit aid; the identity is an elementary polynomial expansion:

```
var('rho sigma iota eta ell n p')
C2 = lambda z: z*(z-1)/2
B = n + p
Delta = p*(2*n + p)
N = n*Delta
A0 = n*p*(2*n + p + 1)/2
R = rho + n*iota + B*eta - n*ell
S = sigma - n*iota - n*eta + B*ell
w = ell - iota - eta
H = lambda X,Y: n*C2(X) + B*X*Y + n*C2(Y)
E = H(rho,sigma) + n*p^2*C2(iota) + n*p*(sigma-rho-n)*iota \
    + (Delta*sigma - A0)*eta \
    + (p*Delta*eta + Delta*rho + A0 - n*p)*ell
expand(E - H(R,S) - N*C2(w+1))
# output: 0
```

Therefore, for fixed e, R, S, equations (A.13)–(A.15) give

$$[q^e(-x)^R(-y)^S]\,\theta^{\mathrm{HM}}_{n,p} = \sum_{w\in\mathbb{Z}} \mathrm{sg}_2\left(R+nw+\frac{n-1}{2}, S-nw-\frac{n+1}{2}\right)\mathbf{1}_{e=H(R,S)+NC_2(w+1)\cdot} \quad \text{(A.16)}$$

Equivalently, since $(-x)^R(-y)^S = (-1)^{R+S}x^Ry^S$,

$$\begin{aligned}[q^e x^R y^S]\,\theta^{\mathrm{HM}}_{n,p} =& (-1)^{R+S}\sum_{w\in\mathbb{Z}} \mathrm{sg}_2\left(R+nw+\frac{n-1}{2}, S-nw-\frac{n+1}{2}\right)\\ &\times \mathbf{1}_{e=H(R,S)+NC_2(w+1)\cdot}\end{aligned} \quad \text{(A.17)}$$

The sum is locally finite: for fixed e, R, S, the equation $e = H(R,S) + NC_2(w+1)$ has at most two integer solutions w.

But the coefficient of $q^e x^R y^S$ in

$$\bar{\Theta}_{n,p} = \sum_{R,S,w\in\mathbb{Z}} (-x)^R(-y)^S\,\mathrm{sg}_2\left(R+nw+\frac{n-1}{2}, S-nw-\frac{n+1}{2}\right) q^{H(R,S)+NC_2(w+1)}$$

is exactly the right-hand side of (A.17). Hence

$$\theta^{\mathrm{HM}}_{n,p}(x,y,q) = \bar{\Theta}_{n,p}(x,y,q)$$

coefficientwise.